



Some tips and tricks

Data Engineer



Lisa
Hoving



Github

Household announcements

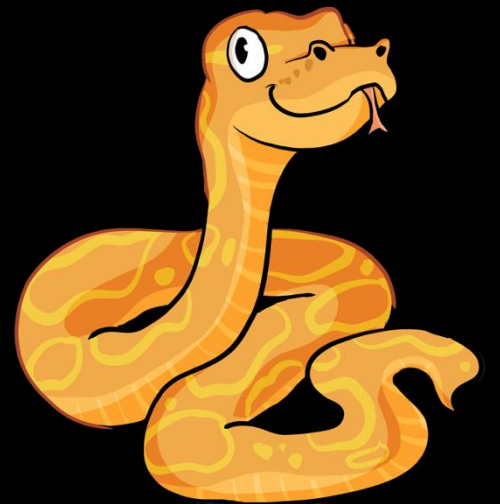
- An overview of options
- Some in preview
 - Need Unity Catalog
- Based on my own experiences

Agenda

- 01** An expensive Python
- 02** Azure Databricks Pricing
- 03** Monitoring & Alerts
- 04** Solutions (1-5)
- 05** Conclusion



1. An expensive python



An expensive python

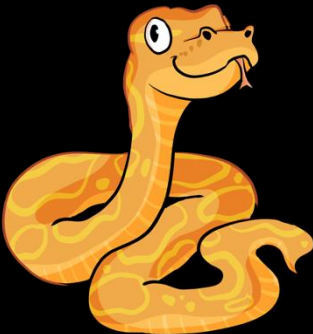
Azure spend

Microsoft Azure Sponsorship

Subscription Cost: \$920.35

SERVICE NAME	SERVICE RESOURCE	SPEND
Storage	Hot GRS Iterative Read Operations	\$387.44
Azure Databricks	Premium All-Purpose Photon DBU	\$374.16
Virtual Machines	D4ds v5	\$102.32
Storage	P15 LRS Disk	\$22.65
Storage	Hot GRS Write Operations	\$19.56
IoT Hub	S1 Unit	\$4.84

(14-06-2024 to 16-06-2024)



2. Azure Databricks Pricing



2. Azure Databricks Pricing

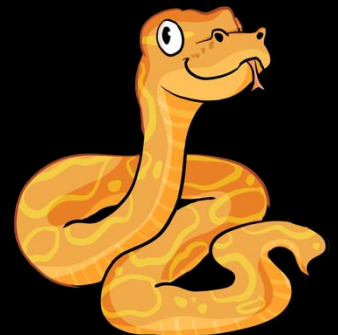
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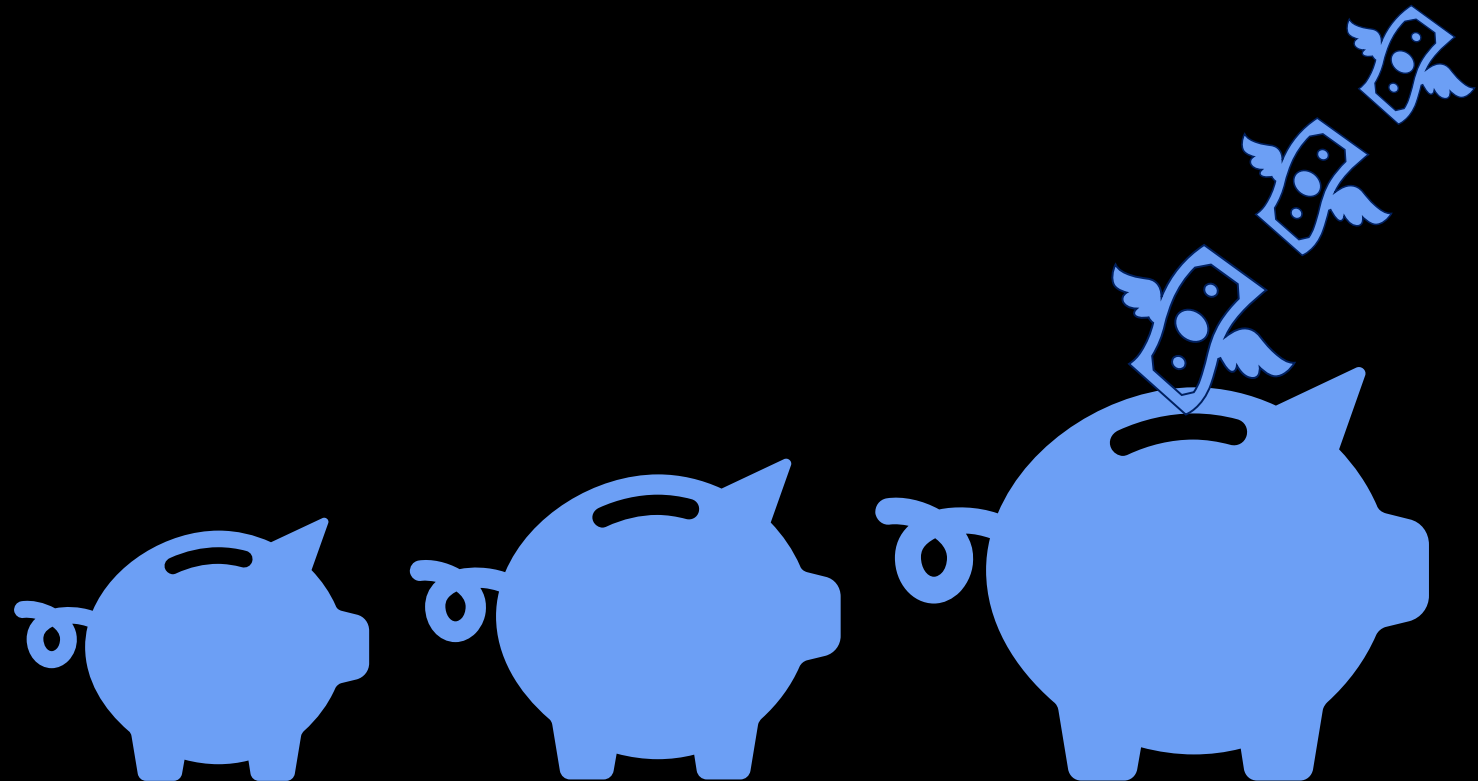
(14-06-2024 to 16-06-2024)



2. Azure Databricks Pricing

Databricks Unit (DBU)

- Normalized unit of processing power
 - Per hour



2. Azure Databricks Pricing

Home >

 **DiscountingData**

Azure Databricks Service

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

 Delete

Essentials

Status

: Active

Resource group

Location

: West Europe

Subscription

Subscription ID

Tags [\(edit\)](#)

: workshop : helpmydatabricksistooexpensive

Managed Resource Group

URL

Pricing Tier

: [Premium \(+ Role-based access controls\)](#) (Click to cha...

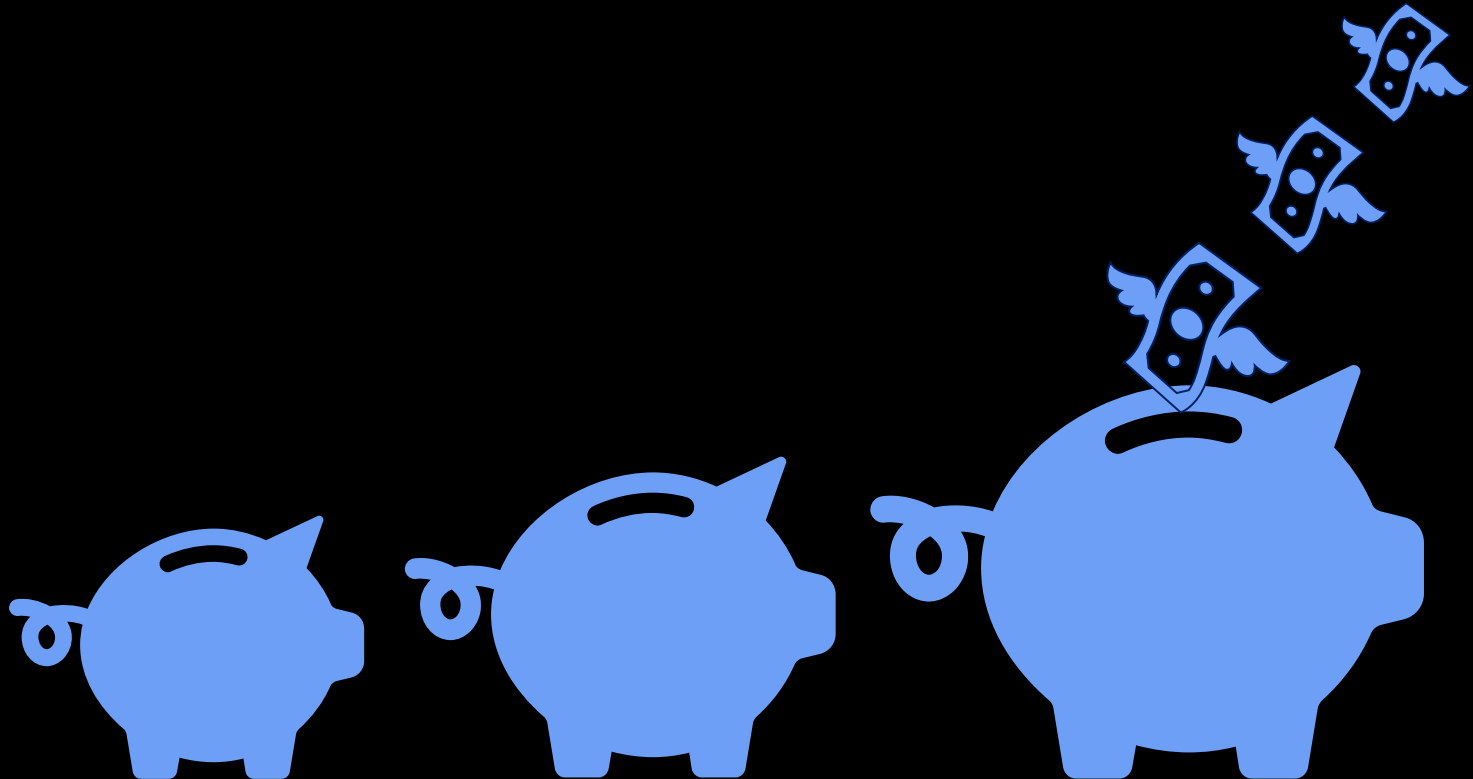
JSON View



Launch Workspace

2. Azure Databricks Pricing

- DBU
- Virtual Machines



2. Azure Databricks Pricing

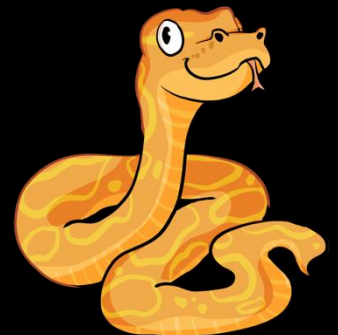
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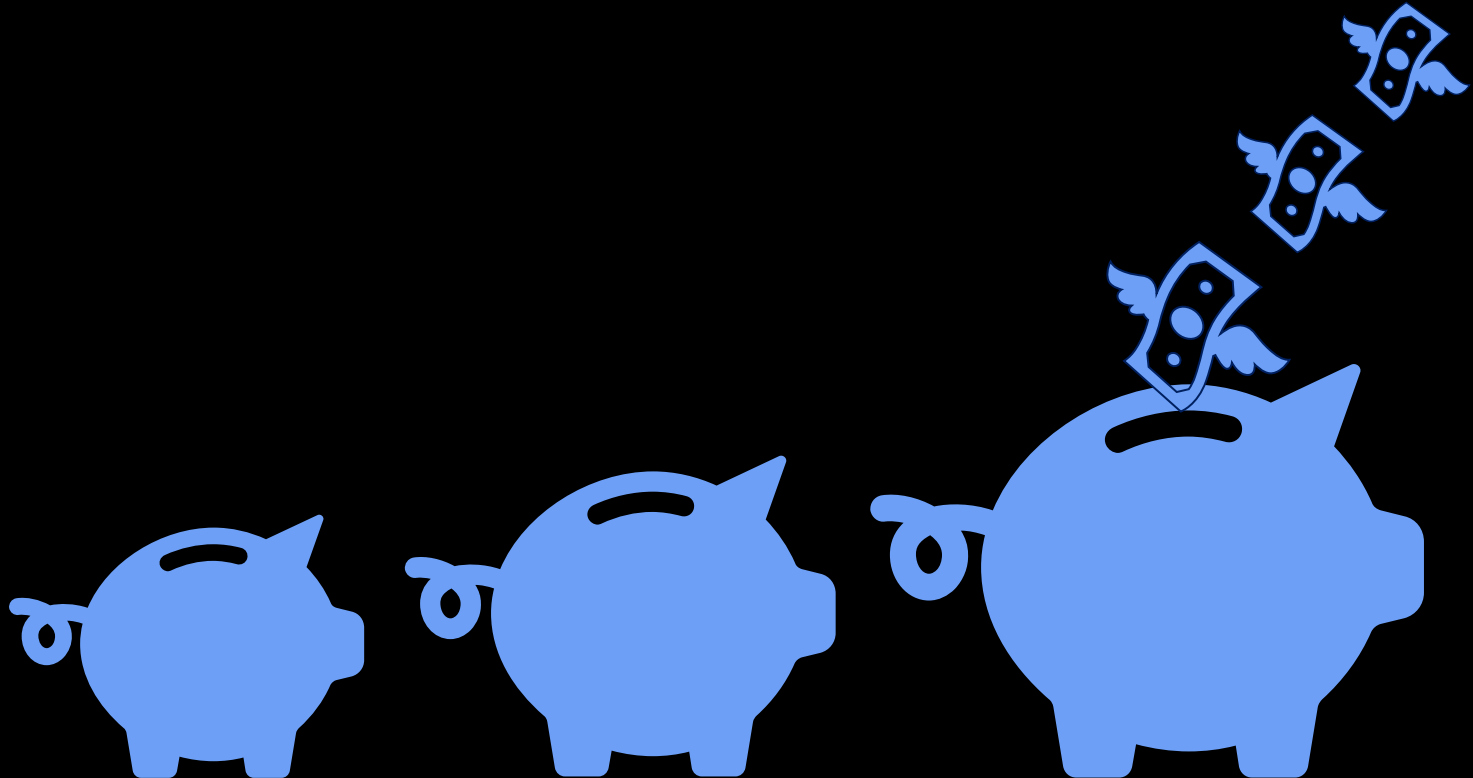
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(14-06-2024 to 16-06-2024)



2. Azure Databricks Pricing

- DBU
- Virtual Machines
 - Disks
 - IP Address
- Other resources
 - Storage Account
 - Key Vault
 - Log Analytics
 - Private Link
 - Data sources



**This is why you
must monitor!**





3. Monitoring & Alerts

3. Monitoring & Alerts

2 Options

- Azure Databricks
(Unity Catalog Public Preview)
 - Workspace Budgets
 - Serverless Budget Policies
- Azure Portal
 - Budgets
 - Alerts



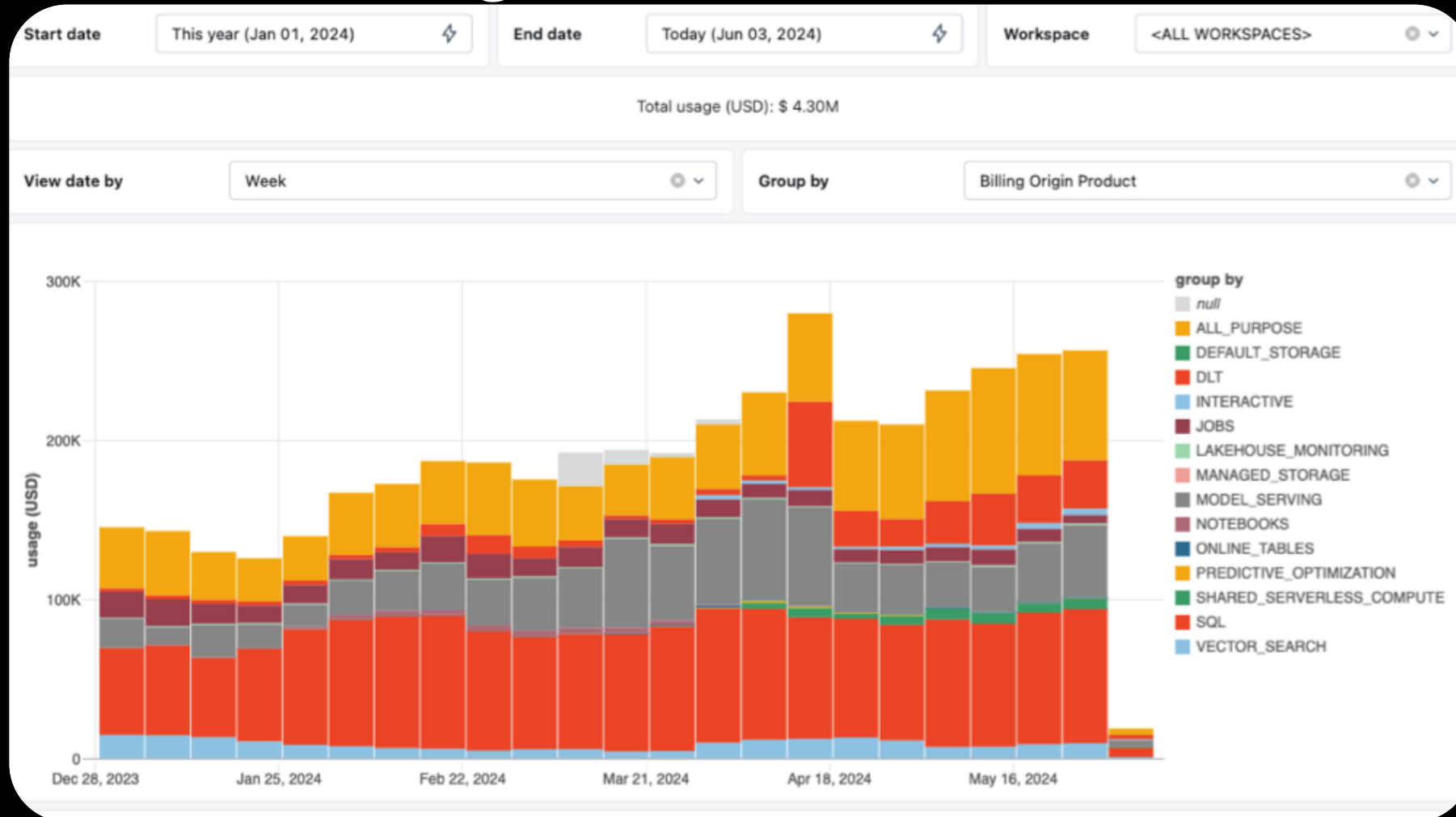
Serverless



Workspace

3. Monitoring & Alerts

Databricks Usage Dashboard



4. Solutions



Solutions

- 01** Optimize Data Source
- 02** Cluster Settings
- 03** Optimize Code
- 04** Make it a Job!
- 05** Stream or Micro Batch?



4. Solutions

(1) Optimize Data Source



4. Solutions

Optimize Data Source

Microsoft Azure Sponsorship

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(14-06-2024 to 16-06-2024)

4. Solutions

Optimize Data Source

- What do your queries cost?
- What techniques are used?
- Can they be more efficient?



Azure cost calculator

West Europe

VM: D4ds_v5

0.27 VM/hour

West Europe

All Purpose Compute (Photon)

Premium Workspace

0.55 DBU/hour

DBU

2/VM

Number of VM's

(workers + driver)

7

Hours

48

Total Cost

VM

\$90.72

DBU

\$369.60

Data Source

\$407.00

Total

\$867.32

Azure cost calculator

West Europe

VM: D4ds_v5

0.27 VM/hour

West Europe

All Purpose Compute (Photon)

Premium Workspace

0.55 DBU/hour

DBU

2/VM

Number of VM's

(workers + driver)

7

Hours

48

Total Cost

VM

\$90.72

DBU

\$369.60

Data Source

\$3.11

Total

\$463.43

47%



4. Solutions

(2) Cluster Settings



4. Solutions

Cluster Settings

Microsoft Azure Sponsorship

Subscription Cost: \$920.35

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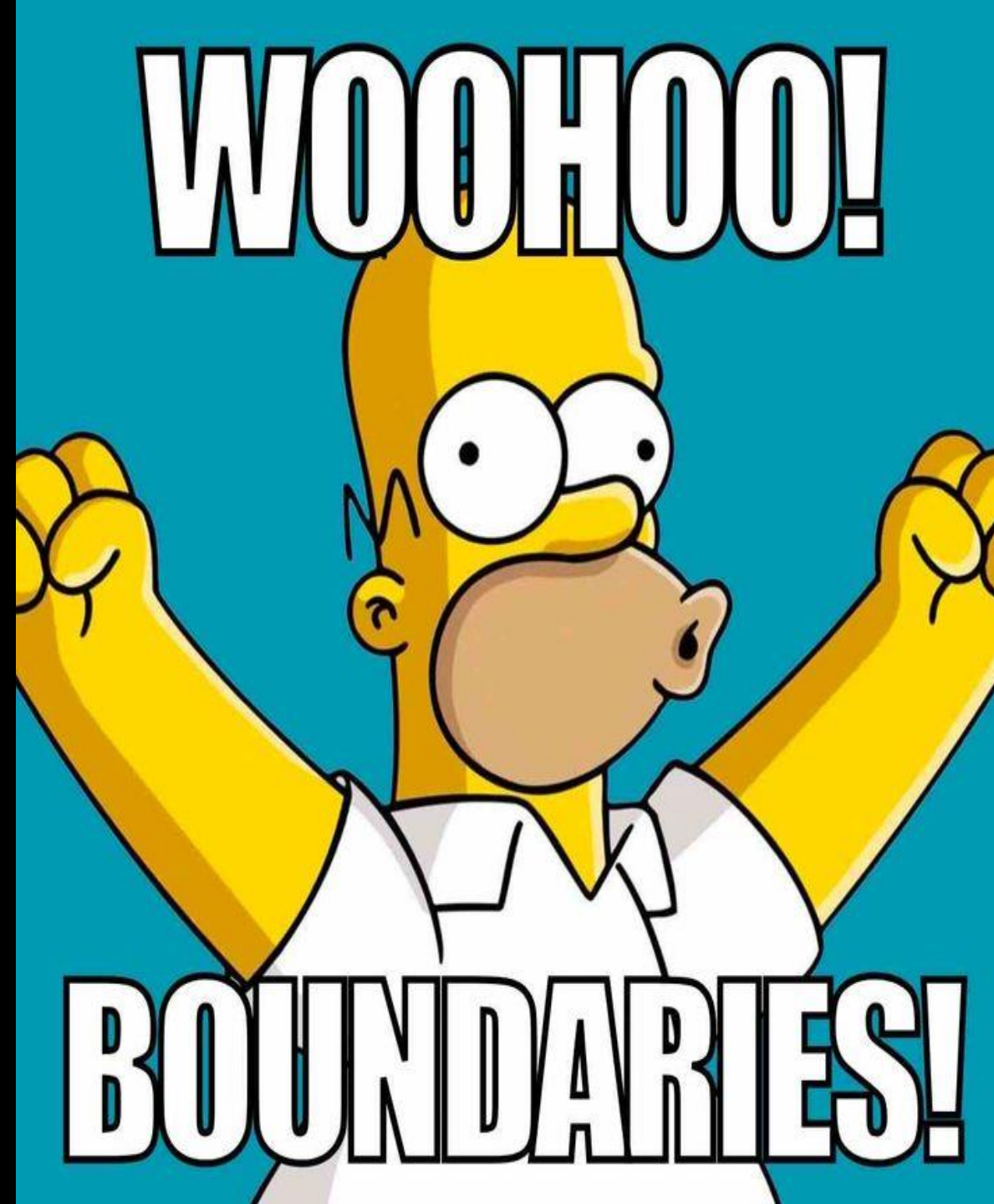
(14-06-2024 to 16-06-2024)

4. Solutions

Cluster Settings

Change DBU

- Photon
- Number of workers (VM's)
- Worker/driver Type

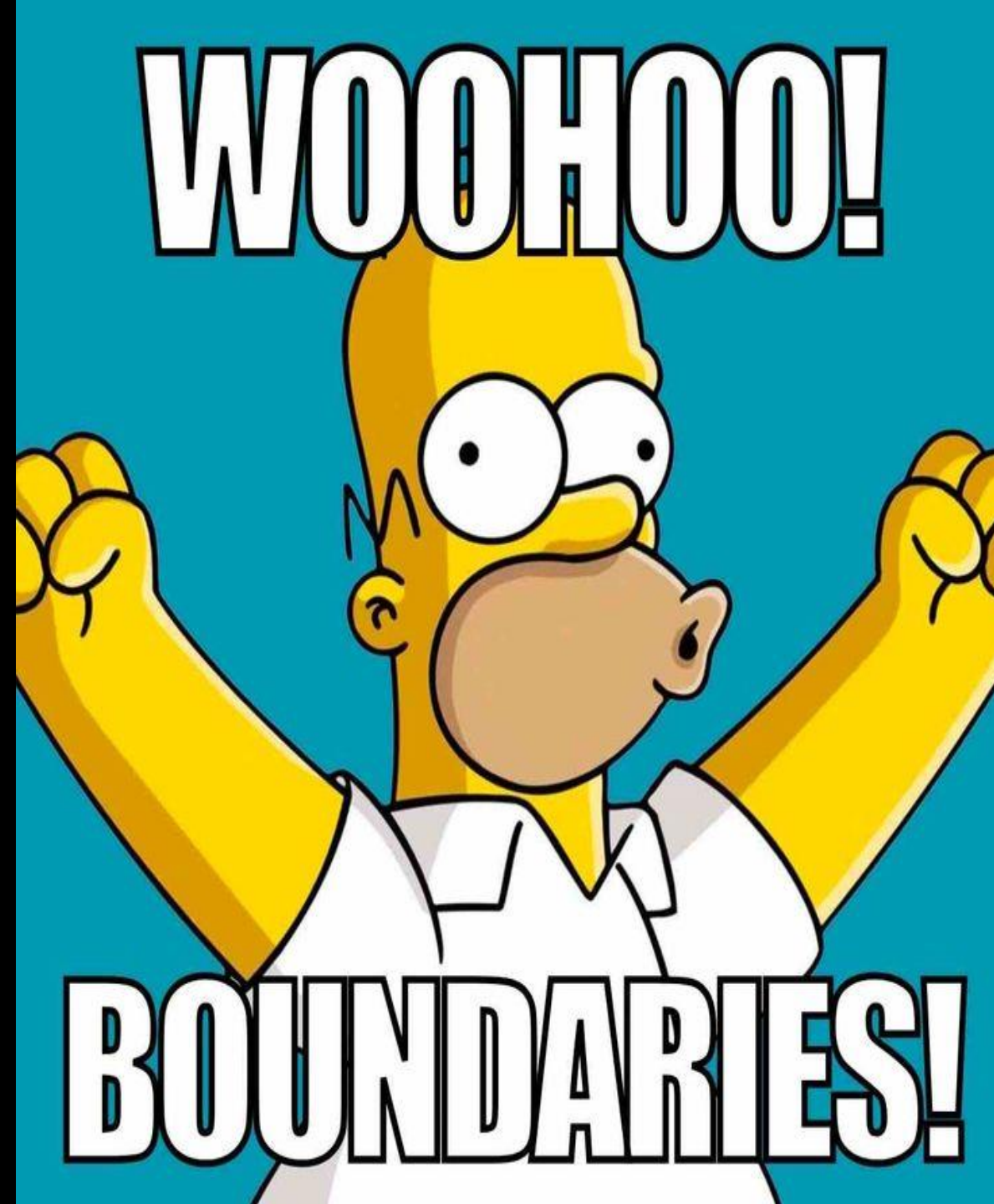


4. Solutions

Cluster Settings

Decrease cluster time

- Auto terminate
- Spark version



4. Solutions

Cluster Settings

Spot instances

- Might decrease price
- Might make workloads
 - Unstable
 - Run longer
- Not for driver nodes



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Hours

48

Total Cost

VM

\$90.72

DBU

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Data Source

\$3.11

Total

\$463.43



Azure cost calculator

West Europe

VM: D4s_v3

0.24 VM/hour

West Europe

All Purpose Compute (no Photon)

Premium Workspace

0.55 DBU/hour

DBU

0.75/VM

Number of VM's

(workers + driver)

2 - 4

Hours

48

Total Cost

VM

\$23.04 - \$46.08

DBU

\$39.60 – \$79.20

Data Source

\$3.11

Total

\$65.75 – \$128.39

72%



4. Solutions

(3) Code Optimization



4. Solutions

Code Optimization

- The most expensive resource? It's-a me!
 - *At some point, you should stop*



4. Solutions

Code Optimization

When?

- Upgrade Apache Spark
- Change UDF to Apache Spark native



4. Solutions

(4) Make it a job!



4. Solutions

Make it a job!

- DBU price differs per workload type
- Jobs compute < All-purpose compute
 - $\$0.30 < \0.55 per DBU/hour



Azure cost calculator

West Europe

VM: D4s_v3

0.24 VM/hour

West Europe

All Purpose Compute (no Photon)

Premium Workspace

0.55 DBU/hour

DBU

0.75/VM

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Hours

48

Total Cost

VM

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Data Source

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Total

\$65.75 – \$128.39

Azure cost calculator

West Europe

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0.24 VM/hour

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Job Compute (no Photon)

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DBU

0.75/VM

Number of VM's

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2 - 4

Hours

48

Total Cost

VM

\$23.04 - \$46.08

DBU

\$21.60 – \$43.20

Data Source

\$3.11

Total

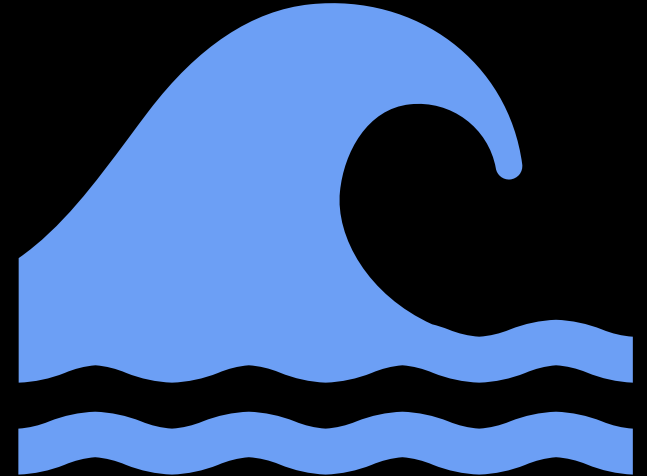
\$47.75 – \$92.39

28%



4. Solutions

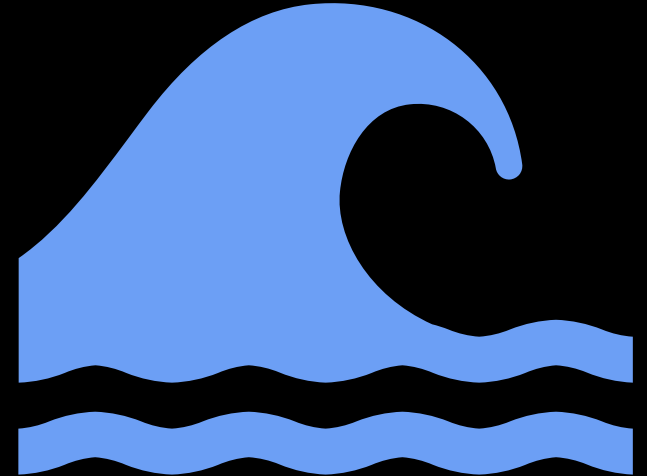
(5) Stream or Micro Batch?



4. Solutions

Stream or Micro Batch?

How real time do you need it to be?



Azure cost calculator

West Europe

VM: D4s_v3

0.24 VM/hour

West Europe

Job Compute (no Photon)

Premium Workspace

0.30 DBU/hour

DBU

0.75/VM

Number of VM's

(workers + driver)

2 - 4

Hours

48

Total Cost

VM

\$23.04 - \$46.08

DBU

\$21.60 – \$43.20

Data Source

\$3.11

Total

\$47.75 – \$92.39

Azure cost calculator

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Job Compute (no Photon)

Premium Workspace

0.30 DBU/hour

DBU

0.75/VM

Number of VM's

(workers + driver)

2 - 4

Hours

24

Total Cost

VM

\$11.52 - \$23.04

DBU

\$10.80 – \$21.60

Data Source

\$3.11

Total

\$25.43 – \$47.75

48%



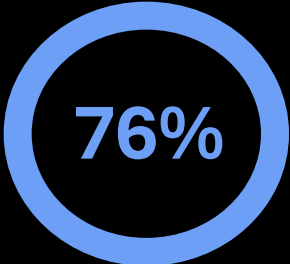
Azure cost calculator

West Europe
Automated Serverless Compute 0.50 DBU/hour

Hours 24

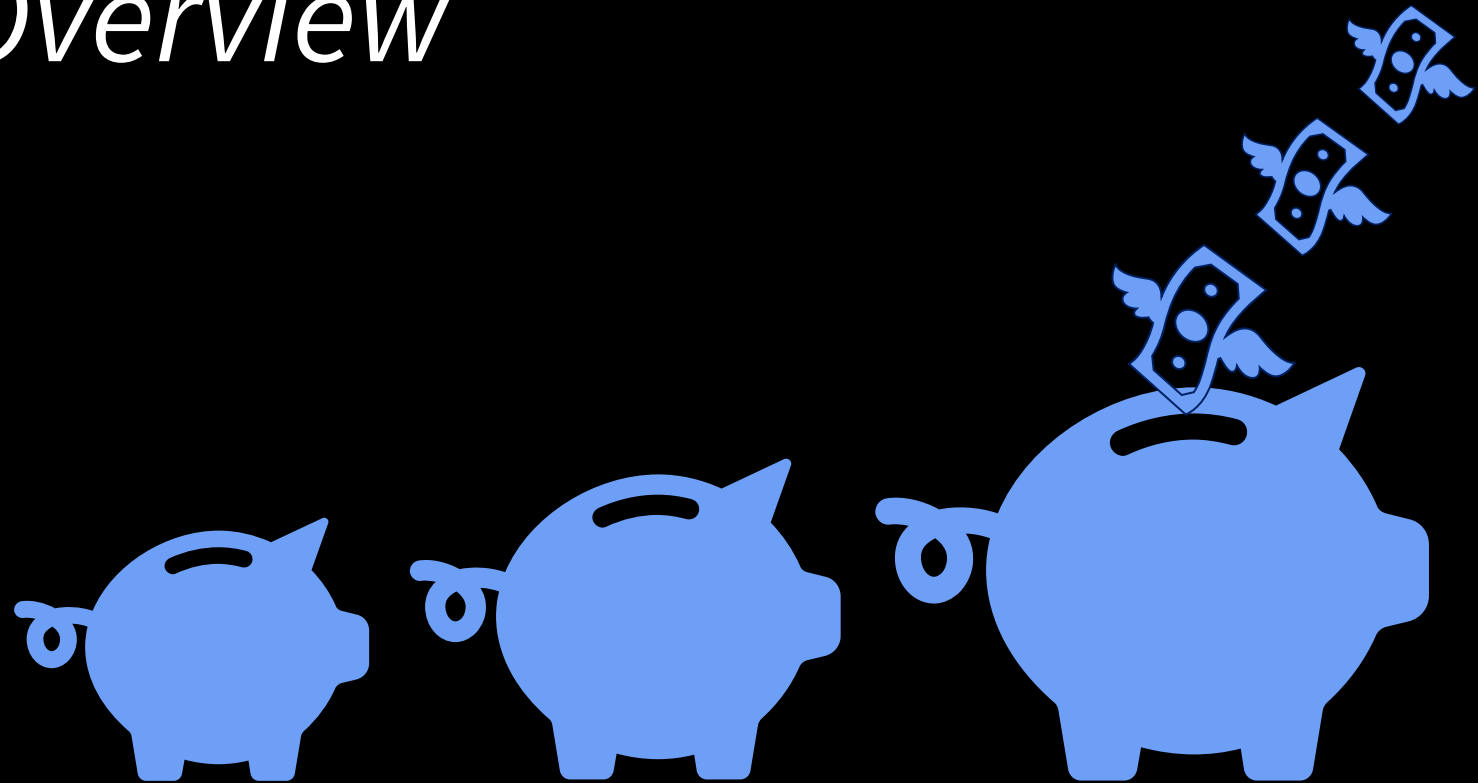
Total Cost

DBU	\$19
Data Source	\$3.11
Total	\$22,11



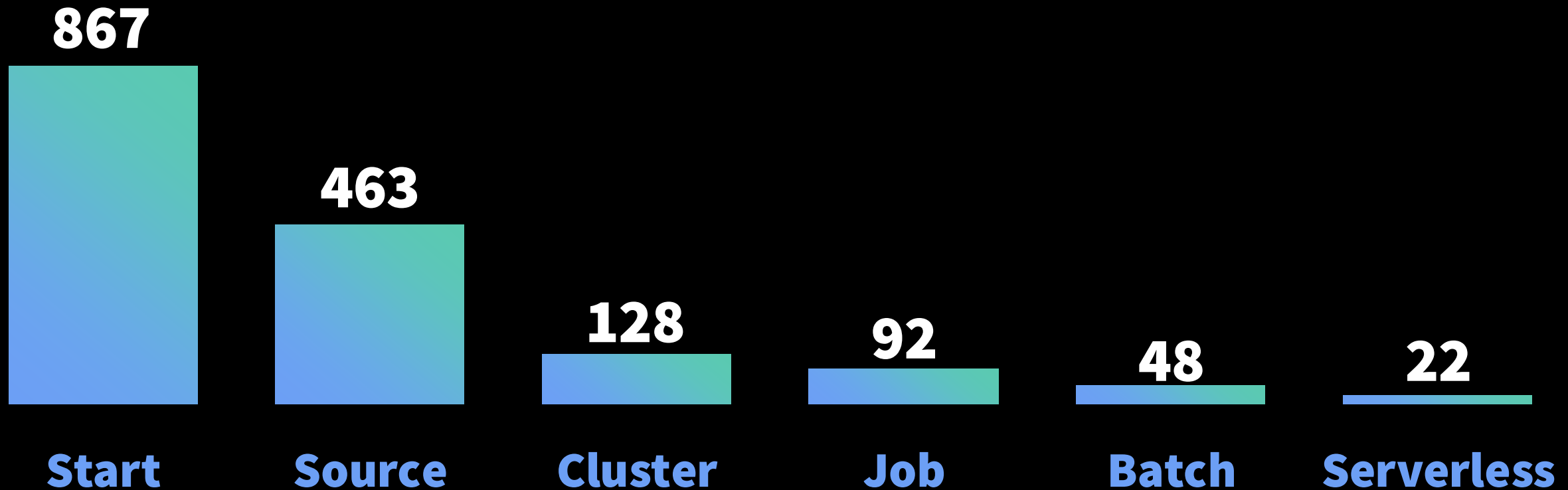
4. Solutions

Overview



4. Solutions

Overview of cost savings

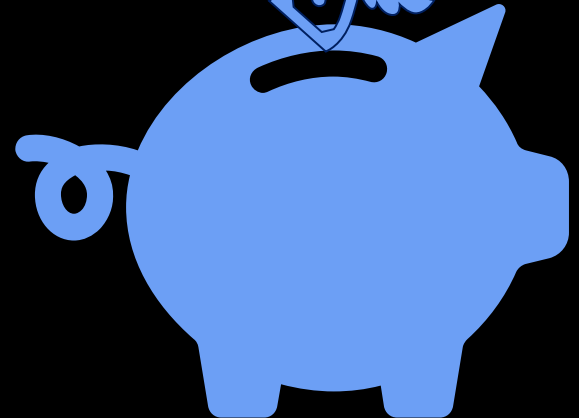
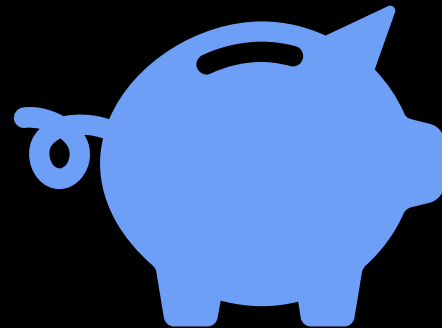
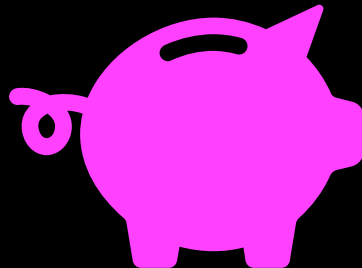
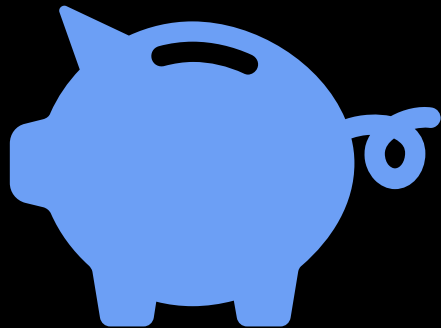
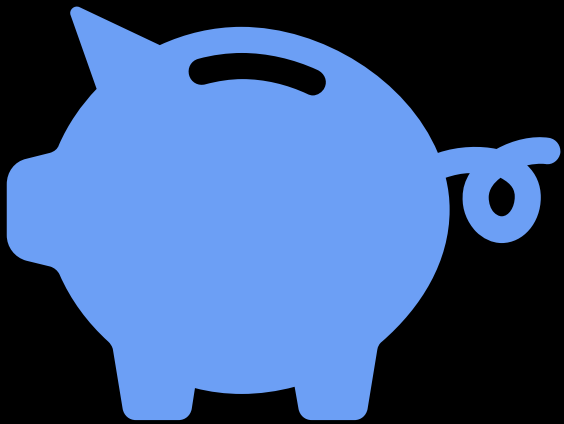


4. Solution

Overview of c

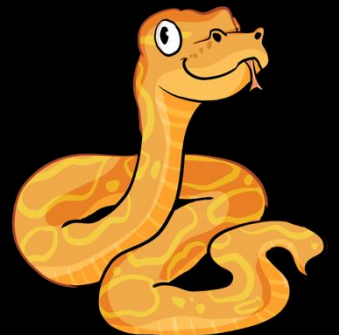


**I could have
saved >\$800**



5. Conclusion

- Total Azure spend
 - DBU
 - VM
 - Data sources
 - Other resources
- Monitoring
 - Alerts: At least in Azure Portal
 - Usage Dashboard: for optimizing costs



5. Conclusion

- Don't forget your Data Sources
- Optimize your cluster settings
- Avoid unnecessary gold-plating of code
- Job clusters are cheaper than General Purpose Compute
- Streaming is expensive

